Review plan · for the teacher

Cleaning and organising data: unit check

Data engineering · Level 2 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- Tidy sheets a computer can use
- Names that agree
- Blanks, zeros and repeats
- One ID per pupil: joining sheets

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet of paper for its answers.
2. Quiz (20 min): Read each question aloud from the quiz sheet; each team agrees on one answer and writes it down.
3. Explain (10 min): Reveal each answer and ask one team to explain why it's right, using an example from the unit.
4. Wrap-up (8 min): Write the question most teams got wrong on the board, and work through it together.

Discussion

- Which sheet in our school would you clean first?
- What would a master list change for us?
- Which check would have caught our worst mistake?

Exit ticket

- One cleaning habit I will use this week, and on which sheet.

Try it at school tomorrow
Take one real sheet your school uses, make a tidy copy with pupil IDs, run the four checks on it and write a short cleaning log.

Quiz sheet · for each team

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Team _____ Date _____

1 **Where do formulas such as AVERAGE and COUNTIF belong?**

- ◯ In a Total row under the pupils
- ◯ On a separate summary sheet that reads the data
- ◯ Typed over the raw marks

2 **Python reads every value in a CSV file as text, so marks need int() before they can be added.**

- ◯ True
- ◯ False

3 **Which of these would a computer count as different from the name "Grace"? Choose all that apply.**

- ☐ " Grace" (with a space at the start)
- ☐ "Grace"
- ☐ "Grayce"
- ☐ "grace"

4 **Put the cleaning steps in order.**

- ☐ Tidy spaces and capitals
- ☐ Match names to the master list
- ☐ Keep the raw sheet untouched
- ☐ Log what you changed
- ☐ Remove true repeats

5 **Four maths marks are 12, 16, blank and 8. What should the report say?**

- ◯ The average is 12, and say nothing about the blank
- ◯ The average is 9, counting the blank as 0
- ◯ The average of 3 marks is 12; 1 mark is missing

6 **Match each check to what it catches.**

In range	• • • •	A mark nobody wrote down
Not empty	• • • •	A mark for pupil 47 in a class of 30
No repeats	• • • •	A mark of 25 out of 20
Known ID	• • • •	The same pupil's mark typed twice

7 **This should print 2, the number of blank marks, but it prints 0. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  marks = ["14", "", "0", ""]
2  blanks = 0
3  for m in marks:
4      if m == "0":
5          blanks = blanks + 1
6  print(blanks)
```

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8 **Why join the register and the mark sheet by pupil ID instead of by name?**

◯ IDs keep the sheet shorter ◯ An ID picks out one pupil only; names repeat and change

◯ Lookups only work with numbers

9 **A mark with no matching pupil should simply be deleted.**

◯ True ◯ False

10 **Fill the gaps: look up each mark's pupil by ID, with ? when there's no match.**

```
names = {"1": "Amina", "2": "Tunde"}
for pupil_id, mark in [("2", 15), ("9", 11)]:
    print(names. _____ (pupil_id,  _____ ), mark)
```

Word bank: `add` `0` `get` `"?"`

Answers · for the teacher

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1 **On a separate summary sheet that reads the data**

The summary sheet reads the tidy data without changing it.

2 **True**

Values from csv.DictReader are text: "14" + "14" is "1414".

3 **"grace"; " Grace" (with a space at the start); "Grayce"**

Only an exact copy matches. Tidying with strip() and lower() fixes the space and the capital; only matching to the class list catches "Grayce".

4 **Top to bottom: 2, 3, 1, 5, 4**

Work on a copy, tidy before matching, and look for repeats only once the names agree.

5 **The average of 3 marks is 12; 1 mark is missing**

Leave the blank out (36 ÷ 3 = 12) and say that one mark is missing.

6 **In range — A mark of 25 out of 20**
Not empty — A mark nobody wrote down
No repeats — The same pupil's mark typed twice
Known ID — A mark for pupil 47 in a class of 30

Each check looks for a different kind of fault.

7 **Line 4: `if m == "":`**

The check looks for "0", a real mark, instead of "", the empty text that a blank cell becomes.

8 **An ID picks out one pupil only; names repeat and change**

A key must pick out exactly one row. Two pupils can share a name and spellings vary, but an ID given once means one pupil.

9 **False**

It's a question to check: an ID typed wrong, or a pupil missing from the list.

10

```
names = {"1": "Amina", "2": "Tunde"}
for pupil_id, mark in [("2", 15), ("9", 11)]:
    print(names.get(pupil_id, "?"), mark)
```

names.get(pupil_id, "?") gives the name for that ID, or "?" when the ID isn't in the dictionary: it prints Tunde 15, then ?